

7-3b Evaluating the Market Equilibrium

Figure 7 shows consumer and producer surplus when a market reaches the equilibrium of supply and demand. Recall that consumer surplus equals the area above the price and under the demand curve and producer surplus equals the area below the price and above the supply curve. Thus, the total area between the supply and demand curves up to the point of equilibrium represents the total surplus in this market.

Figure 7

Consumer and Producer Surplus in the Market Equilibrium

Total surplus—the sum of consumer and producer surplus—is the area between the supply and demand curves up to the equilibrium quantity.

Is this equilibrium allocation of resources efficient? That is, does it maximize total surplus? To answer this question, recall that when a market is in equilibrium, the price determines which buyers and sellers participate in the market. Those buyers who value the good more than the price (represented by the segment AE on the demand curve) choose to buy the good; buyers who value it less than the price (represented by the segment EB) do not. Similarly, those sellers whose costs are less than the price (represented by the segment CE on the supply curve) choose to produce and sell the good; sellers whose costs are greater than the price (represented by the segment ED) do not.

These observations lead to two insights about market outcomes:

1. Free markets allocate the supply of goods to the buyers who value them most, as measured by their willingness to pay.
2. Free markets allocate the demand for goods to the sellers who can produce them at the lowest cost.

Thus, given the quantity produced and sold in a market equilibrium, the social planner cannot increase economic well-being by changing the allocation of consumption among buyers or the allocation of production among sellers.

But can the social planner raise total economic well-being by increasing or decreasing the quantity of the good? The answer is no, as stated in this third insight about market outcomes:

3. Free markets produce the quantity of goods that maximizes the sum of consumer and producer surplus.

Figure 8 illustrates why this is true. To interpret this figure, keep in mind that the demand curve reflects the value to buyers and the supply curve reflects the cost to sellers. At any quantity below the equilibrium level, such as Q_1 , the value to the marginal buyer exceeds the cost to the marginal seller. As a result, increasing the quantity produced and consumed raises total surplus. This continues to be true until the quantity reaches the equilibrium level. Similarly, at any quantity beyond the equilibrium level, such as Q_2 , the value to the marginal buyer is less than the cost to the marginal seller. In this case, decreasing the quantity raises total surplus, and this continues to be true until quantity falls to the equilibrium level. To maximize total surplus, the social planner would choose the quantity at which the supply and demand curves intersect.

Figure 8

The Efficiency of the Equilibrium Quantity

At quantities less than the equilibrium quantity, such as Q_1 , the value to buyers exceeds the cost to sellers. At quantities greater than the equilibrium quantity, such as Q_2 , the cost to sellers exceeds the value to buyers. Therefore, the market equilibrium maximizes the sum of producer and consumer surplus.

Together, these three insights tell us that the market outcome makes the sum of consumer and producer surplus as large as it can be. In other words, the equilibrium outcome is an efficient allocation of resources. The benevolent social planner can, therefore, leave the market outcome just as she finds it. This policy of leaving well enough alone goes by the French expression *laissez-faire*, which literally translates to “leave to do” but is more broadly interpreted as “let people do as they will.”

Society is lucky that the planner doesn't need to intervene. Although it has been a useful exercise imagining what an all-knowing, all-powerful, well-intentioned dictator would do, let's face it: Such characters are hard to come by. Dictators are rarely benevolent, and even if we found someone so virtuous, she would lack crucial information.

Suppose our social planner tried to choose an efficient allocation of resources on her own, instead of relying on market forces. To do so, she would need to know the value of a particular good to every potential consumer in the market and the cost for every potential producer. And she would need this information not only for this market but for every one of the many thousands of markets in the economy. This task is practically impossible, which explains why centrally planned economies never work well.

The planner's job becomes easy, however, once she takes on a partner: Adam Smith's invisible hand of the marketplace. The invisible hand takes all the information about buyers and sellers into account and guides everyone in the market to the best outcome as judged by the standard of economic efficiency. It is a remarkable feat. That is why economists so often advocate free markets as the best way to organize economic activity.

Case Study: Should There Be a Market for Organs?

Some years ago, the front page of *The Boston Globe* ran the headline “How a Mother’s Love Helped Save Two Lives.” The newspaper told the story of Susan Stephens, a woman whose son needed a kidney transplant. When the doctor learned that the mother’s kidney was not compatible, he proposed a novel solution: If Stephens donated one of her kidneys to a stranger, her son would move to the top of the kidney waiting list. The mother accepted the deal, and soon two patients had the transplants they were waiting for.

The ingenuity of the doctor’s proposal and the nobility of the mother’s act cannot be doubted. But the story raises some intriguing questions. If the mother could trade a kidney for a kidney, would the hospital allow her to trade a kidney for an expensive, experimental cancer treatment that she could not otherwise afford? Should she be allowed to exchange her kidney for free tuition for her son at the hospital’s medical school? Should she be able to sell her kidney and use the cash to trade in her old Chevy for a new Lexus?

As a matter of public policy, our society makes it illegal for people to sell their organs. In essence, in the market for organs, the government has imposed a price ceiling of zero. The result, as with any binding price ceiling, is a shortage of the good. The deal in the Stephens case did not fall under this prohibition because no cash changed hands.

Many economists believe that allowing a free market for organs would yield large benefits. People are born with two kidneys, but they usually need only one. Meanwhile, some people suffer from illnesses that leave them without any working kidney. Despite the obvious gains from trade, the current situation is dire: The typical patient has to wait several years for a kidney transplant, and every year thousands of people die because a compatible kidney cannot be found. If those needing a kidney were allowed to buy one from those who have two, the price would rise to balance supply and demand. Sellers would be better off with the extra cash in their pockets. Buyers would be better off with the organ they need to save their lives. The shortage of kidneys would disappear.

Such a market would lead to an efficient allocation of resources, but critics of this plan worry about fairness. A market for organs, they argue, would benefit the rich at the expense of the poor because organs would then be allocated to those most willing and able to pay. But you can also question the fairness of the current system. Now, most of us walk around with an extra organ that we don’t really need, while some of our fellow citizens are dying to get one. Is that fair?

Ask the Experts

Supplying Kidneys

“A market that allows payment for human kidneys should be established on a trial basis to help extend the lives of patients with kidney disease.”

Source: IGM Economic Experts Panel, March 11, 2014

QuickQuiz

7. Isabelle values her time at **\$60** an hour. She spends **2** hours giving Jayla a massage. Jayla was willing to pay as much as **\$300** for the massage, but they negotiated a price of **\$200**. In this transaction,
- a. consumer surplus is **\$20** larger than producer surplus.
 - b. consumer surplus is **\$40** larger than producer surplus.
 - c. producer surplus is **\$20** larger than consumer surplus.
 - d. producer surplus is **\$40** larger than consumer surplus.
8. An efficient allocation of resources maximizes
- a. consumer surplus.
 - b. producer surplus.
 - c. consumer surplus plus producer surplus.
 - d. consumer surplus minus producer surplus.
9. When a market is in equilibrium, the buyers are those with the ____ willingness to pay and the sellers are those with the ____ costs.
- a. highest; highest
 - b. highest; lowest
 - c. lowest; highest
 - d. lowest; lowest

10. Producing a quantity larger than the equilibrium of supply and demand is inefficient because the marginal buyer's willingness to pay is
- a. negative.
 - b. zero.
 - c. positive but less than the marginal seller's cost.
 - d. positive and greater than the marginal seller's cost.

Chapter 7: Consumers, Producers, and the Efficiency of Markets: 7-3b Evaluating the Market Equilibrium

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